

AI Candidate Ranking System

India Runs x Redrob Challenge

Team: Antigravity

1. The Problem: The Keyword Trap

Recruiters go through hundreds of profiles and still miss the right person.

- Keyword filters can't see what actually matters. A 'Marketing Manager' who lists 10 AI tools might score higher than a true 'ML Engineer' with 3 core tools.
- The JD explicitly demands production experience, not just academic/research backgrounds.
- A 'perfect-on-paper' candidate who hasn't logged in for 6 months is useless to a recruiter.

Our Goal:

- Read the JD and understand what the role genuinely needs.
- Look at the full picture: semantic intent, career history, skills, and behavioral signals.
- Deliver a shortlist that recruiters can implicitly trust.

2. Architecture: Hybrid 5-Layer Model

We built a robust, 5-layer hybrid scoring model that aggregates disparate signals:

Layer 1: Semantic Similarity (35% Weight)

- Uses TF-IDF (bigrams, sublinear TF) to compare the candidate's full career narrative against the JD.
- Captures contextual alignment, rewarding candidates who have actually worked on 'ranking' and 'retrieval'.

Layer 2: Skill Match (25% Weight)

- Taxonomy-based keyword matching with proficiency weighting.
- Incorporates Redrob platform skill assessments and peer endorsements.
- Validates hands-on competence in embeddings, vector DBs, and NLP.

3. Architecture: Experience & Signals

Layer 3: Career Trajectory (15% Weight)

- Analyzes ML/AI tenure specifically at product companies.
- Hunts for 'production' keywords (deployed, shipped, scale).
- Penalizes pure consulting backgrounds, aligning strictly with JD constraints.

Layer 4: Experience & Seniority (15% Weight)

- Sweet-spot targeting: Peak scores at 5-9 years of experience.
- Boosts for 'Senior/Lead/Staff' titles; penalizes Junior/Intern.

Layer 5: Behavioral Signals (10% Weight)

- Measures genuine availability: Open to work, short notice periods.

- Factors in recruiter response rates, interview completion, and recent platform activity.

4. JD-Driven Disqualifier Penalties

To prevent gaming the system, we deduct up to 45 points for severe red flags:

- BOLD:Non-ML Current Title (-22 pts)

Accountants or Civil Engineers won't slip through just by adding 'Python'.

- BOLD:Unreachable Candidates (-18 pts)

Recruiter response rates below 10% indicate the candidate isn't actively engaging.

- BOLD:Platform Inactivity (-12 pts)

Penalizes candidates who haven't logged in for >180 days.

- BOLD:Pure Consulting / No Production Evidence (-20 pts combined)

Filters out candidates lacking shipped/deployed product experience.

5. Results & Validation

Execution Stats:

- Processed 100,000 candidates in 35.1 seconds.
- TF-IDF vocabulary size: 20,000.

Top Pick: CAND_0039754

- Senior Applied Scientist with 16.2 years experience at Meta.
- Core skills: Fine-tuning LLMs, Qdrant, NLP, OpenSearch.
- Immediately available (30d notice) & Active GitHub (78/100).
- Score: 1.0000

Validation:

- Passed official 'validate_submission.py' natively.
- Ranks output cleanly, perfectly sorted, with human-readable recruiter reasoning.